Why the old large-cell Γ scheme is not a reference for 2D GW: the spurious charge of fitted pair densities

Ritaj Tyagi, University of Regensburg

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Summary. The large-cell Γ-point GW implementations in CP2K (`tensor_large_cell_Gamma`, `ri_rs_large_cell_Gamma`) build the screened interaction of a 2D supercell on a k-mesh `KPOINTS_W` $n\,n\,1$ of the supercell's small Brillouin zone, with a second "extrapolation" mesh of $2n$ points and a Coulomb matrix $V(\mathbf{q})$ from a minimum-image truncated real-space lattice sum. Band gaps from this scheme were used as the reference against which the new Γ-only scheme (one real k-point plus an analytic head-and-wing correction) was judged, and the two differ by 0.12 eV on the largest cell we can afford. This note documents that the old scheme carries the systematic error, that the error is the spurious net charge of RI-fitted pair densities identified by J. Wilhelm in cp2k PR #6447, and that the Γ-only numbers are confirmed by his full-k-point code to about 15 meV. Numbers that came out of the old scheme for 2D systems should therefore not be used as references.

1 The observation

Monolayer h-BN, $a = 2.502$ Å, 15 Å vacuum, PBE, aug-SZV-MOLOPT-GTH tier-1 orbital and RI bases, 10 minimax points, `CUTOFF_RADIUS_RI` 5 Å, identical in every code and run. "Gap" is the direct G_0W_0 gap at the k-point $P = (11/32, 11/32)$ of the primitive zone, the point nearest to K that an 8×8 supercell with a 2×2 DOS mesh samples.

scheme	cell / mesh	gap (eV)
old large-cell scheme, `KPOINTS_W 4 4 1` + extrapolation	8×8 supercell	8.152
old large-cell scheme, `KPOINTS_W 4 4 1` + extrapolation	14×14 supercell	6.582
old large-cell scheme, `KPOINTS_W 2 2 1` + extrapolation	14×14 supercell	6.595
Γ-only + head/wing correction (this work)	8×8 supercell	6.710
Γ-only + head/wing correction	10×10 supercell	6.705
Γ-only + head/wing correction	14×14 supercell	6.699
full-k RI-RS GW, PR #6447 (primitive cell)	6×6 mesh	6.590
full-k RI-RS GW, PR #6447	8×8 mesh	6.658
full-k RI-RS GW, PR #6447	12×12 mesh	6.681
full-k RI-RS GW, PR #6447	16×16 mesh	6.684
full-k RI-RS GW, PR #6447	20×20 mesh	6.687
full-k RI-RS GW, PR #6447	24×24 mesh	6.690

Three things stand out. (i) The old scheme is not converged with the cell: 8.15 eV at 8×8 against 6.58 eV at 14×14, and an earlier sweep with the tier-2 basis (11.85, 9.74, 9.66, 9.43, 8.63, 7.54 eV for $n = 5, 6, 7, 8, 10, 12$) was still falling at 30 Å. (ii) The Γ-only scheme is converged to 5 meV in the gap from 8×8 on. (iii) The independent full-k-point code converges from below onto the Γ-only value; state by state at the supercell Γ point the absolute G_0W_0 energies agree to 5 meV (valence) and 14 meV (conduction) with the 24×24 mesh, with identical Σ^x and v^{xc} and Σ^c within 10 meV. The old scheme sits 0.12 eV below both at 14×14 and is still drifting downwards, so the 0.12 eV is its error, not the new code's.

2 The mechanism: a spurious charge in the fitted pair densities

In the real-space RI (RI-RS) a product of two orbitals is interpolated from its values on the atom-centred grid points $\mathbf{r}_\ell$ and fitted to the auxiliary basis through the fit coefficients $Z_{\ell P}$,

$$\psi_i(\mathbf{r})\psi_a(\mathbf{r}) \approx \sum_P \Big[\sum_\ell Z_{\ell P}\, \psi_i(\mathbf{r}_\ell)\psi_a(\mathbf{r}_\ell) \Big] \varphi_P(\mathbf{r}) \equiv \sum_P d_{ia,P}\, \varphi_P(\mathbf{r}). \tag{1}$$

The exact transition density is neutral, $\int \psi_i\psi_a = 0$ by orthogonality. The fitted one is not: its charge $m_{ia} = \sum_P c_P\, d_{ia,P}$, with $c_P = \int \varphi_P$, is a small but nonzero number set by the grid and the fit (the grid cannot represent orthogonality, and the fit is regularised). This was found by J. Wilhelm while converging his full-k-point RI-RS GW (cp2k PR #6447, commit `a2a9da4c91`).

A spurious charge is harmless for a molecule and nearly harmless for a 3D crystal, but not in two dimensions. On the $\mathbf{q}$ mesh of a 2D calculation the Coulomb kernel in the charge channel is $v(\mathbf{q}) = 2\pi/(Aq)$, so the charge mode of χ enters the dielectric function as

$$\varepsilon_{00}(\mathbf{q}) - 1 \;\sim\; \frac{2\pi}{Aq} \sum_{ia} \frac{2\,|m_{ia}|^2\,\Delta_{ia}}{\Delta_{ia}^2 + \omega^2} \;=\; \frac{C}{q}, \tag{2}$$

a $1/q$ divergence with a spurious prefactor. Summed over a 2D mesh the mesh points approach $\mathbf{q} = 0$ as $1/N$, and $\sum_{\mathbf{q}} 1/q$ over an $N \times N$ mesh grows like $\ln N$: the screening, and with it the gap, drifts logarithmically with the mesh instead of converging. His numbers: the WSe_2 gap moved from 2.222 to 2.197 eV with the mesh; after projecting the charge out of every fitted density, $d' = d - c\,(c^\dagger d)/(c^\dagger c)$, it is 2.233 eV and stable; the 2D h-BN regression reference moved by +0.92 eV. The fix is in his PR.

The old large-cell scheme has the same fitted densities with the same spurious charge, but its Coulomb matrix is not $2\pi/(Aq)$. $V(\mathbf{q})$ is a real-space lattice sum truncated at the minimum-image radius $R_c \approx L/2$ of the supercell, whose charge-channel value is finite, $\int_0^{R_c} 2\pi r\,\mathrm{d}r/r = 2\pi R_c$ per cell area. The leak therefore does not diverge; it produces a finite spurious screening proportional to $|m|^2 R_c/A \propto 1/L$, i.e. a uniform downward shift of Σ^c that decays only as the inverse cell size. That is exactly the signature in the table: a gap that is too small, converges slowly from below, and is still moving at 30 Å. The sign of Wilhelm's fix (+0.92 eV for h-BN with his settings) and the sign of our discrepancy (old scheme below) agree.

3 Why the Γ-only scheme is immune

The Γ-only implementation never lets a fitted density interact through the charge channel of a sampled $V(\mathbf{q})$:

- the body of the dielectric matrix is built with the neutral projector, $V_b = \Pi V(0)\Pi$, $\Pi = 1 - \hat{c}\hat{c}^{\mathrm{T}}$, so the charge direction of the auxiliary basis carries no Coulomb interaction at all in the body;
- the head and wings of ε in the charge channel are built from the exact velocity-gauge transition dipoles $\langle a|[H,\mathbf{r}]|i\rangle/(\epsilon_a - \epsilon_i)$, checked by the Thomas–Reiche–Kuhn sum rule, and integrated analytically over the whole small Brillouin zone with the $2\pi/q$ kernel;
- the regular $W(\Gamma)$ uses $V(0)$ with only the divergent $G = 0$ constant removed, and the charge direction of that matrix is quenched in ε^{-1} (its Cholesky factorisation drops the negative mode).

This is the same treatment of the charge channel as Wilhelm's Γ model, which is why the two codes agree once his leak is removed. The agreement (15 meV at 8×8 versus 24×24, with the remaining difference of the size expected from the 8×8 cell) is the actual validation of the Γ-only scheme; the old large-cell numbers are not.

4 Consequences

1. Band gaps of 2D systems produced by `tensor_large_cell_Gamma` or `ri_rs_large_cell_Gamma` with `KPOINTS_W` $n\,n\,1$ ($n > 1$) carry a cell-size-dependent error of order 0.1 eV at 30 Å and more than 1 eV on cells of 20 Å, and should not be used as references or regression values.

2. The 0.12 eV difference between the two large-cell schemes at 14×14 is the old scheme's residual leak, consistent with the independent full-k-point result.

3. For 2D the large-cell code should be run Γ-only (`KPOINTS_W 1 1 1`), which activates the head-and-wing correction; the same now holds for 3D cells (`PERIODIC XYZ`), where $V(0)$ is an Ewald sum with the $G = 0$ term omitted.

Data. All inputs and outputs are in `large_cell_gamma/tests/881` of the branch `rirs_periodic_gamma` (`github.com/RitajTyagi/cp2k`); the full-k reference runs and the state-by-state table are in `881/kpoint_ref_pr6447/RESULTS.md`.